

CyberShield SOC Dashboard

A lightweight, rule-based Security Operations Center (SOC) dashboard built as a 45-day university internship project. It lets you upload security log files, automatically detect suspicious activity using predefined rules (no AI/ML), explore the results on an interactive dashboard, and export a PDF incident report.

Project Overview

Organisations generate thousands of security events every day from servers, applications and network devices. Security analysts cannot manually inspect every log line, which makes it easy to miss suspicious activity such as brute-force attacks, repeated login failures, or web scanning.

CyberShield is a small SOC dashboard that collects log files, analyses them with a set of clear detection rules, and presents the findings through an interactive web interface. All detection is rule-based, similar to many real-world SOC setups - no artificial intelligence or machine learning is used.

Objectives

- Provide a simple, centralised place to analyse security logs.
- Parse Linux, Apache and Windows log files into a single structured format.
- Detect suspicious activity using predefined security rules.
- Present alerts with clear severity levels.
- Visualise security events through interactive charts.
- Generate a professional incident report with recommendations.

Features

- Upload log files through the browser (Linux, Apache, Windows).
- Automatic log-type detection based on file content.
- Parsing into a common Pandas DataFrame (one schema for all log types).
- Rule-based threat detection with Low / Medium / High / Critical severities.
- Interactive dashboard: summary metrics, charts, searchable tables and filters.
- One-click PDF incident report generation.
- Friendly error handling for invalid or malformed files.

Technologies Used

Component	Technology
Language	Python
Web UI	Streamlit
Data processing	Pandas
Charts	Plotly Express
PDF reports	ReportLab

Folder Structure

```
CyberShield\SOC\Dashboard/
├── app.py                \# Streamlit entry point (ties the modules together)
├── config.py            \# app settings and folder paths
├── requirements.txt
├── README.md
├── .gitignore
├── modules/
│   ├── parser.py        \# log validation, detection and parsing
│   ├── detector.py      \# rule-based threat detection
│   ├── dashboard.py     \# metrics, charts, tables and filters
│   └── report\_generator.py \# PDF incident report (ReportLab)
├── utils/
│   ├── helpers.py       \# timestamp helpers
│   └── constants.py     \# severity levels, log types, event schema
├── data/
│   ├── sample\_logs/    \# example log files for testing
│   │   ├── sample\_linux.log
│   │   ├── sample\_apache.log
│   │   └── sample\_windows.log
│   └── reports/         \# generated PDF reports are saved here
└── assets/              \# images and other static files
```

Installation

```
\# 1. (Recommended) create a virtual environment
python -m venv venv
source venv/bin/activate          \# on Windows: venv\Scripts\activate

\# 2. Install the dependencies
pip install -r requirements.txt
```

How to Run

```
streamlit run app.py
```

The app opens in your browser at <http://localhost:8501>. Upload one of the files in `data/sample\_logs/` to see it in action.

Sample Log Files

Three sample files are provided in `data/sample\_logs/`. They are designed so that every detection rule is triggered at least once.

File	Type	What it demonstrates
<code>sample_linux.log</code>	Linux	Brute-force login attempts from one IP against <code>bob</code>
<code>sample_apache.log</code>	Apache	A web scanner causing many 403 / 404 / 500 errors
<code>sample_windows.log</code>	Windows	Repeated failed logins on <code>administrator</code> + a spike

Detection Rules

#	Rule	Triggered when	Severity
1	Multiple failed logins from IP	more than 5 failed logins from one IP	High
2	Multiple failed logins for user	more than 5 failed logins for one username	Medium
3	Suspicious repeated IP	one IP appears 10 or more times	Medium
4	High number of HTTP errors	403 > 5, 404 > 5, or 500 > 3	Low / Medium / High
5	Authentication failure spike	failed logins make up more than 60% of auth	Critical

The thresholds are defined at the top of `modules/detector.py` and are easy to change.

Dashboard Features

- **Summary metrics** - total events, total alerts, critical/high alerts, and failed login events.
- **Charts** (Plotly):
 - Events over time (line chart)
 - Alert severity distribution (bar chart)
 - Top source IP addresses (horizontal bar chart)
 - Event type distribution (pie chart)
 - Login status: success vs failed (bar chart)

- **Data tables** - searchable, sortable tables for both parsed logs and alerts.
- **Filters** - sidebar filters for severity, event type, username and source IP that update the whole dashboard automatically.

Report Generation

Clicking **Generate PDF Report** creates a clean PDF (using ReportLab) and saves it in `data/reports/`. The report contains:

1. Cover page (project title, date/time, log file name)
2. Executive summary with an automatically written paragraph
3. Log statistics table
4. Detected incidents table
5. Security recommendations based on the alerts that fired
6. Footer on every page

A download button is also provided in the app.

Limitations

- Works on uploaded files only - there is no live/real-time log monitoring.
- Windows logs must be provided as a simplified CSV export, not raw `.evtx`.
- Detection is rule-based, so it only finds patterns the rules describe.
- No user accounts or authentication - it is a single-user local tool.
- Large log files are held fully in memory (fine for typical coursework sizes).

Future Enhancements

- Live log monitoring and email alerts.
- Optional SQLite storage to keep a history of past analyses.
- Support for more log formats (firewall logs, raw Windows Event Logs).
- User authentication and role-based access.
- Cloud deployment.

Screenshots:

Controls

Upload Log File

Upload

200MB per file • LOG, TXT, CSV

Supported: Linux auth, Apache access, and Windows CSV logs.

CyberShield SOC Dashboard

A lightweight, rule-based Security Operations Center (SOC) dashboard. Upload security log files, detect suspicious activity using predefined rules, view the results, and export an incident report.

Upload a log file from the sidebar to begin.

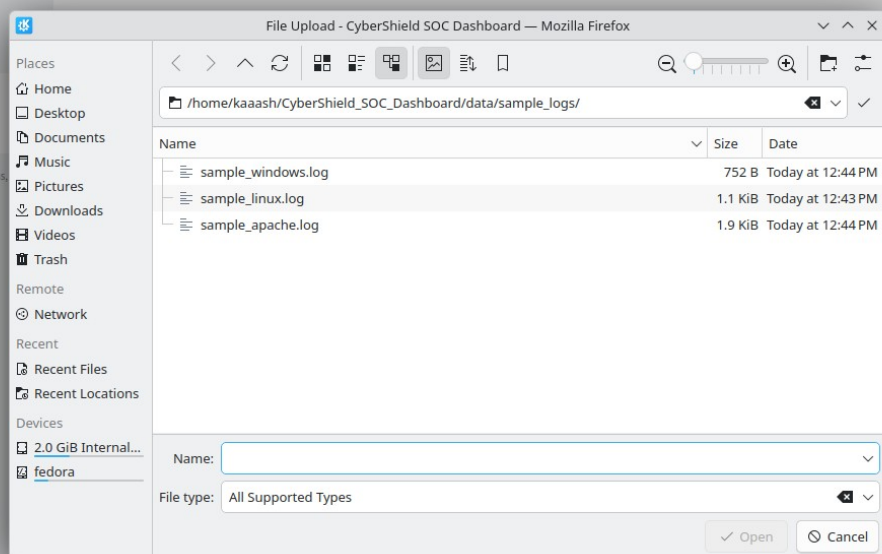
Controls

Upload Log File

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✓	2026-07-24 20:12:19	2026-07-24 20:12:19	5000	failed_login	10000	2026-07-24 20:12:19,10000,5000,2026-07-24 20:12:19,10000	sample_windows.log

▼ Detected Alerts

Search alerts

	Timestamp	Severity	Rule Triggered	Source IP	Username	Description
0	2026-07-24 23:15:19	Critical	Authentication failure spike	-	-	9 of 13 authentication events failed (69%).
1	2026-07-24 09:05:37	High	Multiple failed logins from IP	198.51.100.7	-	6 failed login attempts from IP 198.51.100.7.
2	2026-07-24 09:05:37	Medium	Multiple failed logins for user	-	administrator	6 failed login attempts for user 'administrator'.

Incident Report

The report always covers the full log file, not the current dashboard filters.

Generate PDF Report

Incident Report

The report always covers the full log file, not the current dashboard filters.

Generate PDF Report

Report generated and saved to: /home/kaaash/CyberShield_SOC_Dashboard/data/reports/report_sample_windows_20260724_193607.pdf

Download PDF Report



[Show all downloads](#)

CyberShield SOC Dashboard

Security Incident Report

Report generated: 2026-07-24 19:36

Log file analysed: sample_windows.log

Executive Summary

Metric	Count
Total Events	13
Total Alerts	3
Critical Alerts	1
High Alerts	1
Medium Alerts	1
Low Alerts	0

This report summarises the automated analysis of the uploaded log file. A total of 13 events were parsed and 3 security alerts were generated by the rule-based detection engine. These findings include 1 Critical, 1 High, 1 Medium and 0 Low severity alerts. The presence of high-priority alerts may indicate attacks such as brute-force attempts or scanning, and should be investigated promptly.

Log Statistics

Statistic	Value
Log Type	Windows
Events Parsed	13
Failed Logins	9
Successful Logins	4
Unique Source IPs	7
Unique Usernames	7

Detected Incidents

Timestamp	Severity	Rule Triggered	Source IP	Username	Description
2026-07-24 23:15	Critical	Authentication failure spike	-	-	9 of 13 authentication events failed (69%).
2026-07-24 09:05	High	Multiple failed logins from IP	198.51.100.7	-	6 failed login attempts from IP 198.51.100.7.
2026-07-24 09:05	Medium	Multiple failed logins for user	-	administrator	6 failed login attempts for user 'administrator'.

Security Recommendations

- Strengthen password policies for the targeted user accounts.
- Enable account lockout policies to slow down brute-force attempts.
- Investigate the repeated failed logins from the same IP address.

- Investigate the high proportion of failed authentications across accounts.
- Strengthen password policies.

Author

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- **Course / Programm:** Btech CSE
- **Internship Project:** CyberShield SOC Dashboard (45-day internship)